

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1671

Jet 138 kV
April 02, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1671. The Interconnection Customer proposes a 95.3 MW solar photovoltaic facility interconnecting at the Jet 138 kV substation in the Virginia area. The request was received on February 16, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 575 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,378,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Huntingdon District-Cleburne 138 kV Line	Poland Rd-Jet 138 kV	120.5	232	59.9
Poland Rd-Jet 138 kV	Cleburne-Huntingdon District 138 kV	115.3	244	66.6
Greenway Substation-Cleburne 138 kV	Jet-Poland Rd 138 kV	108.9	226	59.9
Pasco Cnty Solid Waste Resource Recovery-Vmea Peaking Gen 138 kV	Pasco Cnty Solid Waste Resource Recovery-Novec Paradise 138 kV	102.0	365	8.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Huntingdon District-Cleburne 138 kV line	Rebuild line with higher-capacity conductor	\$2,305,000	54% (\$1,245,000)
Jet-Poland Rd 138 kV line	Rebuild line with higher-capacity conductor	\$1,317,000	56% (\$739,000)
Greenway Substation-Cleburne 138 kV	Reconductor line	\$606,000	54% (\$327,000)
Pasco Cnty Solid Waste Resource Recovery-Vmea Peaking Gen 138 kV	Reconductor line	\$1,000,000	7% (\$67,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$2,378,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.